



Advancements In Selective Solder

ADVANCEMENTS	DESCRIPTION	KEY FEATURES	BENEFITS
Electromagnetic Pump Technology	Replaces mechanical pumps with electromagnetic systems for precise solder flow control, minimising dross and maintenance	Stable solder temperature, modular nozzles, turbulence-free delivery, multi-nozzle configurations for simultaneous soldering	Reduced maintenance (up to 50%), consistent wave height, energy efficiency, improved solder joint quality for fine-pitch boards
Multi-Station and Parallel Processing	Dual or multi-station designs allowing simultaneous fluxing, preheating, and soldering on multiple boards or zones	Dual solder pots, parallel processing, adaptable configurations for high-volume or mixed production, robotic automation	Doubled throughput, reduced cycle time (2-3 seconds per assembly), flexibility for high-mix production, lower operational costs
Modular and Compact Design	Scalable, modular systems for easy customisation and space efficiency, suitable for entry-level to high-volume setups	Compact footprint, tool-free maintenance, inline/offline modes, quick nozzle changes, adaptable for lead-free soldering	ROI for small to mid-size lines, adaptability to varying production demands, reduced downtime with 'on the fly' maintenance
AI-Driven Optimization and Real-Time Monitoring	Integration of AI for adaptive soldering, real-time feedback, defect detection, and process optimisation	High-resolution optics, closed-loop flux systems, video recording for traceability, AI-adjusted parameters (speed, temperature)	Improved first-pass yield, reduced operator skill dependency, enhanced traceability with 40-day footage storage, dynamic parameter adjustment for complex boards
Laser and Non-Contact Soldering	Laser-based heating for targeted soldering without physical contact, reducing thermal stress on components	Precise heat control, no nozzles required, compatibility with fine-pitch and heat-sensitive parts, integration with robotic arms	Minimal thermal damage, suitable for miniaturised boards, reduced voids/bridging, eco-friendly with less material use
Green and Lead-Free Technologies	Eco-friendly designs with lead-free solder compatibility, recyclable flux, and energy-efficient operation	Low-dross electromagnetic pumps, nitrogen inerting, reduced flux consumption, compliance with environmental regulations	Lower carbon emissions, reduced waste, cost savings on materials, adaptability to RoHS-compliant production
Closed-Loop Flux and Preheating Systems	Advanced flux dispensers and preheating zones for uniform application and temperature control, minimising defects	Real-time feedback, multi-axis motion control, preheating with infrared or convection, low-residue flux	Better solder joint quality, reduced oxidation, adaptability for complex mixed-technology boards, higher first-pass yields
Energy-Efficient and Miniaturised Systems	Compact, low-energy designs with automated nozzle cleaning and optimised power usage for sustainable manufacturing	Small footprint, tool-free maintenance, N2 heating, digital monitoring for flux/solder flow	Lower operational costs, suitable for space-constrained facilities, compliance with energy regulations, reduced environmental impact

KYORITSU ELECTRIC INDIAWebsite: kyoritsulectric.com

Address: Plot No.10(P), Survey No.234,
235 & 245, India Land Global Industrial
Park, Rajiv Gandhi Infotech Park, Phase 1,
Hinjewadi, Pune, Maharashtra -411057
Email: sales@kyoritsulectric.com
Contact: 9552597468

Inline Brands Supplied: KSM**LEAPTECH CORPORATION**Website: leaptech.in

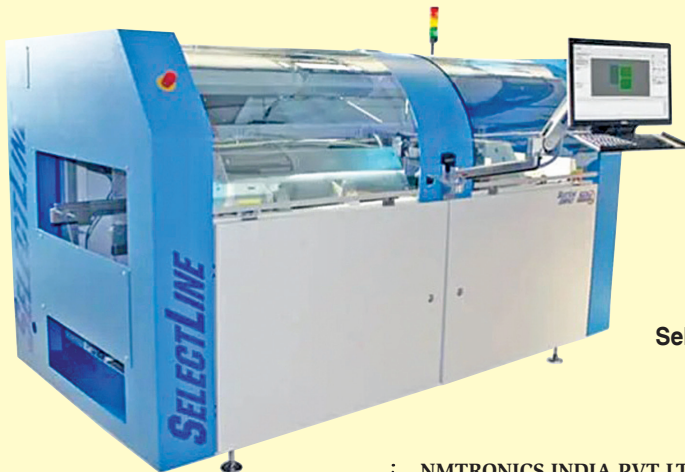
Address: Unit No.240, Raheja Tesla
Industrial, TTC Industrial Area, MIDC
Juinagar, Mumbai, Maharashtra -400705
Email: info@leaptech.in
Contact: 022-20874004/9321710216

Inline Brands Supplied: AST**MECTRONICS MARKETING SERVICES**Website: www.mectronics.in

Address: 313, 2nd Floor, Patparganj
Industrial Area, Functional Industrial
Estate, New Delhi, Delhi -110092
Email: manish@mectronics.in
Contact: 011-43041581

Inline Brands Supplied: AST**Offline Brands Supplied:** EPS**MR TECHNOLOGY**Website: mrtechnology.in

Address: Unit No. 212 & 203, Master Mind I,
Royal Palms, Aarey Colony, Goregaon East,
Mumbai - 400065

**Seho
SelectLine-C**Email: sales@mrtechnology.in

Contact: 9702221244

Offline Brands Supplied: Jaguar**Inline Brands Supplied:** Jaguar**NECTOR EMS TECHNOLOGIES LLP**Website: www.nectorems.com

Address: Street No. 9, Block A,
Dabri Extension East, Dabri, New Delhi,
Delhi -110045

Email: info@nectorems.com

Contact: 9315560437

Inline Brands Supplied: S&M**NMTRONICS INDIA PVT LTD**Website: www.nmtronics.com

Address: SDF No. E-17 & C-2, Noida Special
Economic Zone, Noida Dadri Road, Phase-II,
Noida, Uttar Pradesh -201305

Email: sales@nmtronics.com

Contact: 120-4603500

Inline Brands Supplied: Inertec**NORDSON INDIA PRIVATE LIMITED**Website: www.nordson.com

Address: C-3, #861, Poonamallee
High Road, Kilpauk, Chennai,
Tamil Nadu - 600010

Email: shivakumar.nagarajan@nordson.com

Contact: 9940659762

Inline Brands Supplied: Nordson